

# Lagrangian and Augmented Lagrangian Methods

## 1 Lagrangian and Augmented Lagrangian Methods

### 1.1 Standard Lagrangian Method

#### Problem Formulation

We aim to solve the constrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad c(x) = 0,$$

where:

- $f(x)$  is the objective function,
- $c(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$  represents  $m$  equality constraints.

#### Lagrangian Function

The *Lagrangian function* is defined as:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T c(x),$$

where  $\lambda \in \mathbb{R}^m$  are the Lagrange multipliers.

#### Optimality Conditions (KKT Conditions)

The optimal solution  $(x^*, \lambda^*)$  satisfies the following *Karush-Kuhn-Tucker (KKT) conditions*:

1. **Stationarity:**

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla f(x) + \nabla c(x)^T \lambda = 0,$$

2. **Primal Feasibility:**

$$c(x) = 0.$$

#### Algorithm for the Standard Lagrangian

1. **Initialization:** Start with an initial guess  $x^0$  and  $\lambda^0$ .

2. **Iterative Steps:**

(a) **Primal Update:** Solve the stationarity condition for  $x$ :

$$x^{k+1} = \arg \min_x \mathcal{L}(x, \lambda^k).$$

(b) **Dual Update:** Update the Lagrange multipliers to enforce primal feasibility:

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}),$$

where  $\rho > 0$  is a step size parameter.

3. **Convergence Check:** Stop if:

$$\|\nabla_x \mathcal{L}(x^k, \lambda^k)\| \leq \epsilon \quad \text{and} \quad \|c(x^k)\| \leq \epsilon.$$

## Challenges of the Standard Lagrangian

- Poor convergence near infeasible points.
- Struggles with non-convex problems.
- Feasibility is not strictly enforced during iterations.

## 1.2 Augmented Lagrangian Method (ALM)

### Augmented Lagrangian Function

To address the limitations of the standard method, the *augmented Lagrangian function* adds a quadratic penalty term:

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \lambda^T c(x) + \frac{\rho}{2} \|c(x)\|^2,$$

where  $\rho > 0$  is a penalty parameter.

### Algorithm for ALM

1. **Initialization:** Start with  $x^0$ ,  $\lambda^0$ , and a penalty parameter  $\rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Update:** Solve:

$$x^{k+1} = \arg \min_x \mathcal{L}_\rho(x, \lambda^k).$$

(b) **Dual Update:** Update the Lagrange multipliers:

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}).$$

(c) **Penalty Parameter Update (Optional):** Increase  $\rho$  to enforce stricter feasibility:

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Convergence Check:** Stop if:

$$\|c(x^k)\| \leq \epsilon \quad \text{and} \quad \|\nabla_x \mathcal{L}_\rho(x^k, \lambda^k)\| \leq \epsilon.$$

### Why the Dual Update Takes This Form

The dual update:

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}),$$

is derived from *gradient ascent* on the dual problem:

$$\max_{\lambda} \min_x \mathcal{L}_\rho(x, \lambda).$$

- The gradient of the dual objective is:

$$\nabla_{\lambda} \mathcal{L}_\rho(x, \lambda) = c(x).$$

- Applying gradient ascent on  $\lambda$ :

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}),$$

ensures that  $c(x) \rightarrow 0$  over iterations, gradually enforcing primal feasibility.

### Advantages of ALM

- Combines dual variables and penalty terms for stability.
- Enforces feasibility more robustly.
- Handles non-convex problems better than the standard method.

## 2 Augmented Lagrangian Method for Inequality Constraints

### 2.1 Problem Formulation

We consider the constrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad h(x) \leq 0,$$

where:

- $f(x)$  is the objective function,
- $h(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$  represents  $m$  inequality constraints.

### 2.2 Standard Lagrangian for Inequality Constraints

#### Lagrangian Function

The *Lagrangian function* is defined as:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T h(x),$$

where  $\lambda \in \mathbb{R}_+^m$  are the non-negative Lagrange multipliers.

#### KKT Conditions

The optimal solution  $(x^*, \lambda^*)$  satisfies the following *Karush-Kuhn-Tucker (KKT) conditions*:

1. **Stationarity:**

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla f(x) + \nabla h(x)^T \lambda = 0,$$

2. **Primal Feasibility:**

$$h(x) \leq 0,$$

3. **Dual Feasibility:**

$$\lambda \geq 0,$$

4. **Complementary Slackness:**

$$\lambda_i h_i(x) = 0, \quad \forall i = 1, \dots, m.$$

### 2.3 Augmented Lagrangian for Inequality Constraints

#### Augmented Lagrangian Function

To address the limitations of the standard Lagrangian method, we define the *augmented Lagrangian function* as:

$$\mathcal{L}_\rho(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i h_i(x) + \frac{\rho}{2} \sum_{i=1}^m \max(0, h_i(x))^2,$$

where  $\rho > 0$  is the penalty parameter.

#### Explanation of Terms

- **Lagrangian Term:** The term  $\lambda^T h(x)$  enforces dual feasibility ( $\lambda \geq 0$ ).
- **Penalty Term:** The term  $\frac{\rho}{2} \max(0, h_i(x))^2$  penalizes infeasible solutions ( $h_i(x) > 0$ ).

## 2.4 Iterative Algorithm for Augmented Lagrangian

1. **Initialization:** Start with  $x^0$ ,  $\lambda^0$ , and a penalty parameter  $\rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Update:** Solve:

$$x^{k+1} = \arg \min_x \mathcal{L}_\rho(x, \lambda^k).$$

(b) **Dual Update:** Update the Lagrange multipliers:

$$\lambda_i^{k+1} = \max(0, \lambda_i^k + \rho h_i(x^{k+1})), \quad \forall i = 1, \dots, m.$$

(c) **Penalty Parameter Update (Optional):** Optionally increase  $\rho$  to enforce stricter feasibility:

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Convergence Check:** Stop if:

$$\|h(x^k)\| \leq \epsilon, \quad \|\nabla_x \mathcal{L}_\rho(x^k, \lambda^k)\| \leq \epsilon,$$

and  $\lambda^k$  satisfies complementary slackness.

## 2.5 Why the Dual Update Takes This Form

The dual update:

$$\lambda_i^{k+1} = \max(0, \lambda_i^k + \rho h_i(x^{k+1})),$$

is derived as follows:

- The gradient of the dual objective with respect to  $\lambda_i$  is:

$$\nabla_{\lambda_i} \mathcal{L}_\rho(x, \lambda) = h_i(x).$$

- The dual update:

$$\lambda_i^{k+1} = \lambda_i^k + \rho h_i(x^{k+1}),$$

moves  $\lambda_i$  in the direction of  $h_i(x)$ .

- The  $\max(0, \cdot)$  operation ensures that  $\lambda_i \geq 0$ , maintaining dual feasibility.

## 2.6 Applications of Generalized ALM

- Optimization problems with inequality constraints.
- Machine learning tasks with regularization or constraints.
- Optimal control problems with state or input bounds.

# 3 Augmented Lagrangian Method for Equality and Inequality Constraints

## 3.1 Problem Formulation

We consider the constrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad c(x) = 0, \quad h(x) \leq 0,$$

where:

- $f(x)$  is the objective function,
- $c(x) : \mathbb{R}^n \rightarrow \mathbb{R}^p$  represents  $p$  equality constraints,
- $h(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$  represents  $m$  inequality constraints.

### 3.2 Augmented Lagrangian Function

The *augmented Lagrangian function* for this problem is defined as:

$$\mathcal{L}_\rho(x, \lambda, \mu) = f(x) + \lambda^T c(x) + \mu^T h(x) + \frac{\rho}{2} \|c(x)\|^2 + \frac{\rho}{2} \sum_{i=1}^m \max(0, h_i(x))^2,$$

where:

- $\lambda \in \mathbb{R}^p$  are the Lagrange multipliers for the equality constraints,
- $\mu \in \mathbb{R}_+^m$  are the non-negative Lagrange multipliers for the inequality constraints,
- $\rho > 0$  is the penalty parameter,
- $\max(0, h_i(x))$  enforces penalties only for violated inequality constraints ( $h_i(x) > 0$ ).

### 3.3 Iterative Algorithm

1. **Initialization:** Start with  $x^0$ ,  $\lambda^0$ ,  $\mu^0$ , and a penalty parameter  $\rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Update:** Solve the optimization problem to update  $x$ :

$$x^{k+1} = \arg \min_x \mathcal{L}_\rho(x, \lambda^k, \mu^k).$$

(b) **Dual Update for Equality Constraints:** Update the Lagrange multipliers for  $c(x) = 0$ :

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}).$$

(c) **Dual Update for Inequality Constraints:** Update the Lagrange multipliers for  $h(x) \leq 0$ :

$$\mu_i^{k+1} = \max(0, \mu_i^k + \rho h_i(x^{k+1})), \quad \forall i = 1, \dots, m.$$

(d) **Penalty Parameter Update (Optional):** Optionally increase the penalty parameter  $\rho$  for stricter feasibility:

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Convergence Check:** Stop if the following conditions are satisfied:

$$\|c(x^k)\| \leq \epsilon, \quad \|h_+(x^k)\| \leq \epsilon, \quad \text{and} \quad \|\nabla_x \mathcal{L}_\rho(x^k, \lambda^k, \mu^k)\| \leq \epsilon,$$

where  $\epsilon > 0$  is a small tolerance.

### 3.4 Explanation of the Dual Updates

- **Equality Constraints:** The update for  $\lambda$  is:

$$\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}),$$

which ensures that the equality constraint  $c(x) = 0$  is enforced over iterations. The term  $\rho c(x)$  penalizes violations of the equality constraints.

- **Inequality Constraints:** The update for  $\mu$  is:

$$\mu_i^{k+1} = \max(0, \mu_i^k + \rho h_i(x^{k+1})).$$

This ensures that:

- $\mu_i \geq 0$  (dual feasibility),
- $\mu_i h_i(x) = 0$  (complementary slackness).

The  $\max(0, \cdot)$  operation automatically disregards constraints that are already satisfied ( $h_i(x) \leq 0$ ).

### 3.5 Advantages of the Generalized ALM

- Combines the benefits of dual variables and penalty terms, providing stability and robustness.
- Handles both equality and inequality constraints in a unified framework.
- Enforces feasibility for constraints more effectively compared to standard methods.

### 3.6 Applications of the Generalized ALM

- **Optimization Problems:** General-purpose constrained optimization problems.
- **Machine Learning:** Problems with regularization or constraints, such as fairness-aware optimization or constrained learning.
- **Control Systems:** Solving optimal control problems with state and input constraints.

## 4 Comparison of Penalty Method and Augmented Lagrangian Method

### 4.1 Problem Formulation

We consider the general constrained optimization problem:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad c(x) = 0, \quad h(x) \leq 0,$$

where:

- $c(x) : \mathbb{R}^n \rightarrow \mathbb{R}^p$  represents  $p$  equality constraints,
- $h(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$  represents  $m$  inequality constraints.

### 4.2 Comparison of Methods

### 4.3 Updates for Each Method

#### 4.3.1 Penalty Method

- **Primal Update:**

$$x^{k+1} = \arg \min_x \left( f(x) + \frac{\rho_k}{2} \|c(x)\|^2 + \frac{\rho_k}{2} \sum_{i=1}^m \max(0, h_i(x))^2 \right).$$

- **Penalty Parameter Update:**

$$\rho_{k+1} = \tau \rho_k, \quad \tau > 1.$$

#### 4.3.2 Augmented Lagrangian Method (ALM)

- **Primal Update:**

$$x^{k+1} = \arg \min_x \left( f(x) + \lambda^T c(x) + \mu^T h(x) + \frac{\rho}{2} \|c(x)\|^2 + \frac{\rho}{2} \sum_{i=1}^m \max(0, h_i(x))^2 \right).$$

- **Dual Updates:**

$$\begin{aligned} \lambda^{k+1} &= \lambda^k + \rho c(x^{k+1}), \\ \mu_i^{k+1} &= \max(0, \mu_i^k + \rho h_i(x^{k+1})), \quad \forall i = 1, \dots, m. \end{aligned}$$

### 4.4 Summary

- Use the **Penalty Method** for simple, small-scale problems where implementation simplicity is preferred, and small constraint violations are acceptable.
- Use **ALM** for complex, large-scale problems requiring faster convergence, better numerical stability, and strict feasibility.

| Aspect                       | Penalty Method                                                                               | Augmented Lagrangian Method (ALM)                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective Function           | $P_\rho(x) = f(x) + \frac{\rho}{2}\ c(x)\ ^2 + \frac{\rho}{2}\sum_{i=1}^m \max(0, h_i(x))^2$ | $\mathcal{L}_\rho(x, \lambda, \mu) = f(x) + \lambda^T c(x) + \mu^T h(x) + \frac{\rho}{2}\ c(x)\ ^2 + \frac{\rho}{2}\sum_{i=1}^m \max(0, h_i(x))^2$ |
| Primal Update                | $x^{k+1} = \arg \min_x P_{\rho_k}(x)$                                                        | $x^{k+1} = \arg \min_x \mathcal{L}_\rho(x, \lambda^k, \mu^k)$                                                                                      |
| Dual Updates                 | Not applicable.                                                                              | $\lambda^{k+1} = \lambda^k + \rho c(x^{k+1}), \quad \mu_i^{k+1} = \max(0, \mu_i^k + \rho h_i(x^{k+1}))$                                            |
| Penalty Parameter ( $\rho$ ) | Requires $\rho \rightarrow \infty$ for strict feasibility, leading to ill-conditioning.      | Moderate $\rho$ suffices, improving numerical stability.                                                                                           |
| Convergence Speed            | Slower, especially for large $\rho$ .                                                        | Faster due to combined penalties and dual variable updates.                                                                                        |
| Numerical Stability          | Poor for large $\rho$ .                                                                      | More stable, even for large-scale or ill-conditioned problems.                                                                                     |
| Implementation               | Simpler to implement, suitable for basic problems.                                           | Slightly more complex, but better for challenging or large-scale problems.                                                                         |
| Applicability                | Works for small-scale problems or where small constraint violations are tolerable.           | Ideal for problems requiring strict feasibility and faster convergence.                                                                            |

Table 1: Comparison of Penalty Method and Augmented Lagrangian Method

## 5 Augmented Lagrangian Method with Block Coordinate Descent

### 5.1 Problem Formulation

We aim to solve the constrained optimization problem:

$$\min_{x,y} f(x,y) \quad \text{s.t.} \quad h(x) \leq 0, \quad k(y) \leq 0,$$

where:

- $f(x,y)$  is the objective function,
- $h(x) \leq 0$ : inequality constraints dependent on  $x$ ,
- $k(y) \leq 0$ : inequality constraints dependent on  $y$ .

### 5.2 Augmented Lagrangian Function

The augmented Lagrangian function for this problem is:

$$\mathcal{L}_\rho(x,y,\mu_h,\mu_k) = f(x,y) + \mu_h^T h(x) + \mu_k^T k(y) + \frac{\rho}{2}\|h^+(x)\|^2 + \frac{\rho}{2}\|k^+(y)\|^2,$$

where:

- $\mu_h \in \mathbb{R}_+^m$ : Lagrange multipliers for  $h(x) \leq 0$ ,
- $\mu_k \in \mathbb{R}_+^p$ : Lagrange multipliers for  $k(y) \leq 0$ ,

- $\rho > 0$ : penalty parameter,
- $h^+(x) = \max(0, h(x))$ : positive part of  $h(x)$ ,
- $k^+(y) = \max(0, k(y))$ : positive part of  $k(y)$ .

### 5.3 Algorithm with Block Coordinate Descent

1. **Initialization:** Choose initial values  $x^0, y^0, \mu_h^0, \mu_k^0, \rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Update (BCD):** Alternately update  $x$  and  $y$  by minimizing the augmented Lagrangian:

$$\begin{aligned} x^{k+1} &= \arg \min_x \mathcal{L}_\rho(x, y^k, \mu_h^k, \mu_k^k), \\ y^{k+1} &= \arg \min_y \mathcal{L}_\rho(x^{k+1}, y, \mu_h^k, \mu_k^k). \end{aligned}$$

Specifically:

- **Update for  $x$ :**

$$x^{k+1} = \arg \min_x f(x, y^k) + \mu_h^T h(x) + \frac{\rho}{2} \|h^+(x)\|^2.$$

- **Update for  $y$ :**

$$y^{k+1} = \arg \min_y f(x^{k+1}, y) + \mu_k^T k(y) + \frac{\rho}{2} \|k^+(y)\|^2.$$

(b) **Dual Updates:**

- Update the Lagrange multipliers for  $h(x) \leq 0$ :

$$\mu_{h,i}^{k+1} = \max(0, \mu_{h,i}^k + \rho h_i(x^{k+1})), \quad \forall i.$$

- Update the Lagrange multipliers for  $k(y) \leq 0$ :

$$\mu_{k,j}^{k+1} = \max(0, \mu_{k,j}^k + \rho k_j(y^{k+1})), \quad \forall j.$$

(c) **Penalty Parameter Update (Optional):** Optionally increase  $\rho$ :

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Convergence Check:** Stop if:

$$\|h^+(x^k)\| \leq \epsilon, \quad \|k^+(y^k)\| \leq \epsilon,$$

and:

$$\|\nabla_x \mathcal{L}_\rho(x^k, y^k, \mu_h^k, \mu_k^k)\| \leq \epsilon, \quad \|\nabla_y \mathcal{L}_\rho(x^k, y^k, \mu_h^k, \mu_k^k)\| \leq \epsilon,$$

where  $\epsilon > 0$  is a small tolerance.

### 5.4 Advantages

- Efficient handling of  $x$ - and  $y$ -dependent constraints using block coordinate descent.
- Combines dual variable updates with penalty terms for faster convergence.
- Scalable for large problems where  $x$  and  $y$  have different structures or dimensions.



## 6 Augmented Lagrangian Method with Auxiliary Variables

### 6.1 Problem Reformulation

Given the constrained optimization problem:

$$\min_x f(x) \quad \text{s.t.} \quad c_1(x) = 0, \quad h_1(x) \leq 0, \quad c_2(x) = 0, \quad h_2(x) \leq 0,$$

we introduce an auxiliary variable  $y$  and rewrite the problem as:

$$\begin{aligned} \min_{x,y} f(x) \quad \text{s.t.} \quad & c_1(x) = 0, \quad h_1(x) \leq 0, \\ & c_2(y) = 0, \quad h_2(y) \leq 0, \\ & x - y = 0. \end{aligned}$$

### 6.2 Augmented Lagrangian Function

The augmented Lagrangian for this reformulated problem is:

$$\begin{aligned} \mathcal{L}_\rho(x, y, \lambda_1, \mu_1, \lambda_2, \mu_2, \nu) = & f(x) + \lambda_1^T c_1(x) + \mu_1^T h_1(x) + \frac{\rho}{2} (\|c_1(x)\|^2 + \|h_1^+(x)\|^2) \\ & + \lambda_2^T c_2(y) + \mu_2^T h_2(y) + \frac{\rho}{2} (\|c_2(y)\|^2 + \|h_2^+(y)\|^2) + \nu^T (x - y) + \frac{\rho}{2} \|x - y\|^2, \end{aligned}$$

where:

- $\lambda_1, \lambda_2$ : Lagrange multipliers for equality constraints  $c_1(x) = 0$  and  $c_2(y) = 0$ ,
- $\mu_1, \mu_2$ : Lagrange multipliers for inequality constraints  $h_1(x) \leq 0$  and  $h_2(y) \leq 0$ ,
- $\nu$ : Lagrange multipliers for the coupling constraint  $x - y = 0$ ,
- $\rho > 0$ : penalty parameter.

### 6.3 Algorithm with Block Coordinate Descent

1. **Initialization:** Choose initial values  $x^0, y^0, \lambda_1^0, \mu_1^0, \lambda_2^0, \mu_2^0, \nu^0, \rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Update (Block Coordinate Descent):** Alternately update  $x$  and  $y$  by minimizing the augmented Lagrangian:

$$x^{k+1} = \arg \min_x \mathcal{L}_\rho(x, y^k, \lambda_1^k, \mu_1^k, \lambda_2^k, \mu_2^k, \nu^k),$$

$$y^{k+1} = \arg \min_y \mathcal{L}_\rho(x^{k+1}, y, \lambda_1^k, \mu_1^k, \lambda_2^k, \mu_2^k, \nu^k).$$

Specifically:

- **Update for  $x$ :**

$$x^{k+1} = \arg \min_x f(x) + \lambda_1^T c_1(x) + \mu_1^T h_1(x) + \frac{\rho}{2} (\|c_1(x)\|^2 + \|h_1^+(x)\|^2) + \nu^T (x - y^k) + \frac{\rho}{2} \|x - y^k\|^2.$$

- **Update for  $y$ :**

$$y^{k+1} = \arg \min_y \lambda_2^T c_2(y) + \mu_2^T h_2(y) + \frac{\rho}{2} (\|c_2(y)\|^2 + \|h_2^+(y)\|^2) - \nu^T (x^{k+1} - y) + \frac{\rho}{2} \|x^{k+1} - y\|^2.$$

(b) **Dual Updates:**

- Update  $\lambda_1$  and  $\mu_1$  for  $x$ :

$$\lambda_1^{k+1} = \lambda_1^k + \rho c_1(x^{k+1}), \quad \mu_{1,i}^{k+1} = \max(0, \mu_{1,i}^k + \rho h_{1,i}(x^{k+1})), \quad \forall i.$$

- Update  $\lambda_2$  and  $\mu_2$  for  $y$ :

$$\lambda_2^{k+1} = \lambda_2^k + \rho c_2(y^{k+1}), \quad \mu_{2,i}^{k+1} = \max(0, \mu_{2,i}^k + \rho h_{2,i}(y^{k+1})), \quad \forall i.$$

- Update  $\nu$  for the coupling constraint:

$$\nu^{k+1} = \nu^k + \rho(x^{k+1} - y^{k+1}).$$

(c) **Penalty Parameter Update (Optional):** Optionally increase  $\rho$ :

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Convergence Check:** Stop if:

$$\|c_1(x^k)\| \leq \epsilon, \quad \|h_1^+(x^k)\| \leq \epsilon, \quad \|c_2(y^k)\| \leq \epsilon, \quad \|h_2^+(y^k)\| \leq \epsilon,$$

and:

$$\|x^k - y^k\| \leq \epsilon,$$

where  $\epsilon > 0$  is a small tolerance.

## 6.4 Advantages

- Efficient handling of constraints by separating  $x$ - and  $y$ -dependent components.
- Block coordinate descent simplifies the primal update step.
- The use of dual variables ensures robust enforcement of constraints.

## 7 Multi-Task Learning with Star-Shaped Interaction

### 7.1 Problem Description

We consider a multi-task learning framework where:

- Each **user** is associated with a task, having their own unique data and parameters.
- A **shared representation**  $z$  serves as the central parameter to capture common features across tasks.
- Users interact only with the central  $z$ , following a star-shaped structure, and do not communicate with each other.

The goal is to solve:

$$\min_{z, \{w_t\}} \sum_{t=1}^T \mathcal{L}_t(w_t, z) + \lambda R(z) + \mu \sum_{t=1}^T R_t(w_t),$$

subject to:

$$h_1(z) \leq 0, \quad h_2(w_t) \leq 0 \quad \forall t,$$

where:

- $\mathcal{L}_t(w_t, z)$ : Loss for task  $t$ , dependent on shared parameter  $z$  and task-specific parameter  $w_t$ ,
- $R(z)$ : Regularization on the shared parameter  $z$  (e.g., smoothness or sparsity),
- $R_t(w_t)$ : Regularization on task-specific parameters  $w_t$ ,
- $h_1(z) \leq 0$ : Constraints on  $z$ ,
- $h_2(w_t) \leq 0$ : Constraints on  $w_t$ , e.g., positivity or sparsity.

### 7.2 Reformulation with Auxiliary Variables

To decouple the optimization problem, we introduce auxiliary variables  $y_t$  for each task:

$$\min_{z, \{w_t, y_t\}} \sum_{t=1}^T \mathcal{L}_t(w_t, z) + \lambda R(z) + \mu \sum_{t=1}^T R_t(w_t),$$

subject to:

$$h_1(z) \leq 0, \quad h_2(y_t) \leq 0 \quad \forall t, \quad w_t - y_t = 0 \quad \forall t.$$

### 7.3 Augmented Lagrangian Function

The augmented Lagrangian for this reformulated problem is:

$$\mathcal{L}_\rho(z, \{w_t, y_t\}, \lambda, \{\mu_t, \nu_t\}) = \sum_{t=1}^T \mathcal{L}_t(w_t, z) + \lambda^T h_1(z) + \frac{\rho}{2} \|h_1^+(z)\|^2 + \sum_{t=1}^T \left( \mu_t^T h_2(y_t) + \frac{\rho}{2} \|h_2^+(y_t)\|^2 + \nu_t^T (w_t - y_t) + \frac{\rho}{2} \|w_t - y_t\|^2 \right)$$

where:

- $\lambda$ : Dual variable for constraints  $h_1(z) \leq 0$ ,
- $\mu_t$ : Dual variables for task-specific constraints  $h_2(y_t) \leq 0$ ,
- $\nu_t$ : Dual variables for coupling constraints  $w_t - y_t = 0$ ,
- $\rho > 0$ : Penalty parameter for constraint violations.

## 7.4 Iterative Algorithm: ALM with Block Coordinate Descent

1. **Initialization:** Choose  $z^0, w_t^0, y_t^0, \lambda^0, \mu_t^0, \nu_t^0, \rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Updates (Block Coordinate Descent):**

- Update  $z$ :

$$z^{k+1} = \arg \min_z \sum_{t=1}^T \mathcal{L}_t(w_t^k, z) + \lambda^T h_1(z) + \frac{\rho}{2} \|h_1^+(z)\|^2.$$

- For each task  $t$ , update  $w_t$  and  $y_t$ :

$$w_t^{k+1} = \arg \min_{w_t} \mathcal{L}_t(w_t, z^{k+1}) + \nu_t^T (w_t - y_t^k) + \frac{\rho}{2} \|w_t - y_t^k\|^2,$$

$$y_t^{k+1} = \arg \min_{y_t} \mu_t^T h_2(y_t) + \frac{\rho}{2} \|h_2^+(y_t)\|^2 + \nu_t^T (w_t^{k+1} - y_t) + \frac{\rho}{2} \|w_t^{k+1} - y_t\|^2.$$

(b) **Dual Updates:**

- Update  $\lambda$ :

$$\lambda^{k+1} = \max(0, \lambda^k + \rho h_1(z^{k+1})).$$

- Update  $\mu_t$  and  $\nu_t$  for each task  $t$ :

$$\mu_t^{k+1} = \max(0, \mu_t^k + \rho h_2(y_t^{k+1})), \quad \nu_t^{k+1} = \nu_t^k + \rho(w_t^{k+1} - y_t^{k+1}).$$

(c) **Penalty Parameter Update (Optional):**

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Stopping Criterion:** Stop if the constraints  $h_1(z)$ ,  $h_2(y_t)$ , and  $w_t - y_t$  are satisfied within a tolerance.

## 7.5 Privacy and Security Enhancements

- **Star-Shaped Structure:**

- Users only interact with the central  $z$ , preserving privacy by avoiding direct user-to-user communication.
- Each user performs local updates on  $w_t$  and  $y_t$ , ensuring data remains private.

- **Differential Privacy:**

- Add noise to user updates sent to the central node:

$$\nabla_z \mathcal{L}_t^{\text{noisy}} = \nabla_z \mathcal{L}_t + \mathcal{N}(0, \sigma^2),$$

where  $\mathcal{N}(0, \sigma^2)$  is Gaussian noise.

- **Federated Learning:**

- Each user computes updates locally and only shares aggregated gradients or parameters with the central node.

- **Secure Aggregation:**

- Ensure task-specific contributions are aggregated securely to prevent leakage of individual updates.

- **Encrypted Communication:**

- Use secure encryption protocols for communication between users and the central node.

## 7.6 Summary

This method applies ALM with Block Coordinate Descent to solve multi-task learning problems in a star-shaped structure. The framework inherently limits information sharing and, with additional privacy-preserving techniques, ensures robust privacy and security while maintaining effective shared learning across tasks.

## 8 Decentralized Resource Allocation in Smart Grids

### 8.1 Problem Description

In a smart grid system, multiple distributed energy sources (e.g., solar panels, wind turbines, battery storage) collaborate to meet energy demands in a cost-effective manner. Each energy source has its own operational constraints and objectives, while a central controller coordinates the overall resource allocation. This can be formulated as the following constrained optimization problem:

$$\min_{z, \{x_i\}} \sum_{i=1}^N C_i(x_i) + R(z),$$

subject to:

$$\sum_{i=1}^N x_i = z, \quad h_i(x_i) \leq 0, \quad g(z) \leq 0,$$

where:

- $x_i$ : Power output of energy source  $i$ ,
- $C_i(x_i)$ : Cost function for energy production at source  $i$ ,
- $z$ : Aggregated total power output,
- $R(z)$ : Regularization on the aggregated output (e.g., smoothness),
- $h_i(x_i) \leq 0$ : Operational constraints for source  $i$  (e.g., capacity limits),
- $g(z) \leq 0$ : Global constraints on the aggregated output (e.g., grid stability).

### 8.2 Reformulation with Auxiliary Variables

To decouple the coupling constraint  $\sum_{i=1}^N x_i = z$ , we introduce auxiliary variables  $y_i$  and reformulate the problem as:

$$\min_{z, \{x_i, y_i\}} \sum_{i=1}^N C_i(x_i) + R(z),$$

subject to:

$$h_i(x_i) \leq 0, \quad g(z) \leq 0, \quad x_i - y_i = 0, \quad \sum_{i=1}^N y_i = z.$$

### 8.3 Augmented Lagrangian Function

The augmented Lagrangian for the reformulated problem is:

$$\mathcal{L}_\rho(z, \{x_i, y_i\}, \{\mu_i, \nu\}) = \sum_{i=1}^N \left( C_i(x_i) + \mu_i^T h_i(x_i) + \frac{\rho}{2} \|h_i^+(x_i)\|^2 + \nu_i^T (x_i - y_i) + \frac{\rho}{2} \|x_i - y_i\|^2 \right) + R(z) + \nu^T \left( \sum_{i=1}^N y_i - z \right) + \frac{\rho}{2}$$

where:

- $\mu_i$ : Dual variables for  $h_i(x_i) \leq 0$ ,
- $\nu_i$ : Dual variables for  $x_i - y_i = 0$ ,
- $\nu$ : Dual variable for  $\sum_{i=1}^N y_i = z$ ,
- $\rho > 0$ : Penalty parameter for constraint violations.

## 8.4 Iterative Algorithm: ALM with Block Coordinate Descent

1. **Initialization:** Choose  $z^0, x_i^0, y_i^0, \mu_i^0, \nu_i^0, \nu^0, \rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Updates (Block Coordinate Descent):**

- Update  $x_i$  for each energy source  $i$ :

$$x_i^{k+1} = \arg \min_{x_i} C_i(x_i) + \mu_i^T h_i(x_i) + \frac{\rho}{2} \|h_i^+(x_i)\|^2 + \nu_i^T (x_i - y_i^k) + \frac{\rho}{2} \|x_i - y_i^k\|^2.$$

- Update  $y_i$  for each energy source  $i$ :

$$y_i^{k+1} = \arg \min_{y_i} \nu_i^T (x_i^{k+1} - y_i) + \frac{\rho}{2} \|x_i^{k+1} - y_i\|^2 + \nu^T y_i + \frac{\rho}{2} \left\| \sum_{j=1}^N y_j - z^k \right\|^2.$$

- Update  $z$  (global variable):

$$z^{k+1} = \arg \min_z R(z) + \nu^T \left( \sum_{i=1}^N y_i^{k+1} - z \right) + \frac{\rho}{2} \left\| \sum_{i=1}^N y_i^{k+1} - z \right\|^2.$$

(b) **Dual Updates:**

- Update  $\mu_i$  for  $h_i(x_i) \leq 0$ :

$$\mu_i^{k+1} = \max(0, \mu_i^k + \rho h_i(x_i^{k+1})).$$

- Update  $\nu_i$  for  $x_i - y_i = 0$ :

$$\nu_i^{k+1} = \nu_i^k + \rho(x_i^{k+1} - y_i^{k+1}).$$

- Update  $\nu$  for  $\sum_{i=1}^N y_i = z$ :

$$\nu^{k+1} = \nu^k + \rho \left( \sum_{i=1}^N y_i^{k+1} - z^{k+1} \right).$$

(c) **Penalty Parameter Update (Optional):**

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1.$$

3. **Stopping Criterion:** Stop if:

$$\|h_i^+(x_i^k)\| \leq \epsilon, \quad \|x_i^k - y_i^k\| \leq \epsilon, \quad \left\| \sum_{i=1}^N y_i^k - z^k \right\| \leq \epsilon,$$

and the gradient norms are sufficiently small.

## 8.5 Privacy and Security Enhancements

- **Local Updates:** Each energy source  $i$  updates  $x_i$  and  $y_i$  locally using its own data and constraints.
- **Secure Aggregation:** The central node  $z$  receives only aggregated updates  $\sum_{i=1}^N y_i$ , masking individual contributions.
- **Differential Privacy:** Add noise to updates sent to the central node:

$$\nabla_z^{\text{noisy}} = \nabla_z + \mathcal{N}(0, \sigma^2),$$

where  $\mathcal{N}(0, \sigma^2)$  is Gaussian noise.

- **Encryption:** Use secure encryption protocols for all communications between energy sources and the central node.

## 8.6 Summary

This application demonstrates the effectiveness of the ALM with Block Coordinate Descent for decentralized resource allocation in smart grids. The method ensures efficient handling of distributed components while preserving privacy through local updates and secure aggregation, making it suitable for large-scale systems with complex constraints.

# 9 Decentralized Augmented Lagrangian Method with Block Coordinate Descent

## 9.1 Problem Description

We consider a decentralized optimization problem with  $N$  agents. Each agent  $i$  has a local variable  $x_i$ , and the global variable  $x$  is the concatenation of these local variables:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}.$$

The optimization problem is:

$$\min_x f(x) = \sum_{i=1}^N f_i(x_i),$$

subject to:

$$h_i(x_i) \leq 0 \quad \forall i, \quad c_{ij}(x_i, x_j) = 0 \quad \forall (i, j) \in \mathcal{E}.$$

## 9.2 Reformulation with Auxiliary Variables

Introduce auxiliary variables  $y_{ij}$  to decouple the coupling constraints:

$$\min_{\{x_i, y_{ij}\}} \sum_{i=1}^N f_i(x_i),$$

subject to:

$$h_i(x_i) \leq 0, \quad x_i - y_{ij} = 0, \quad y_{ij} - x_j = 0 \quad \forall (i, j) \in \mathcal{E}.$$

## 9.3 Augmented Lagrangian Function

The augmented Lagrangian is:

$$\begin{aligned} \mathcal{L}_\rho(\{x_i, y_{ij}\}, \{\lambda_{ij}, \mu_{ij}\}) &= \sum_{i=1}^N f_i(x_i) + \sum_{i=1}^N \mu_i^T h_i(x_i) + \frac{\rho}{2} \sum_{i=1}^N \|h_i^+(x_i)\|^2 \\ &+ \sum_{(i,j) \in \mathcal{E}} \left( \lambda_{ij}^T (x_i - y_{ij}) + \frac{\rho}{2} \|x_i - y_{ij}\|^2 + \mu_{ij}^T (y_{ij} - x_j) + \frac{\rho}{2} \|y_{ij} - x_j\|^2 \right). \end{aligned}$$

## 9.4 Algorithm: ALM with Block Coordinate Descent

1. **Initialization:** Initialize  $x_i^0, y_{ij}^0, \lambda_{ij}^0, \mu_{ij}^0, \rho > 0$ .

2. **Iterative Steps:**

(a) **Primal Updates (BCD):**

- Update  $x_i$  for each agent:

$$x_i^{k+1} = \arg \min_{x_i} f_i(x_i) + \mu_i^T h_i(x_i) + \frac{\rho}{2} \|h_i^+(x_i)\|^2 + \sum_{j \in \mathcal{N}(i)} \left( \lambda_{ij}^T (x_i - y_{ij}^k) + \frac{\rho}{2} \|x_i - y_{ij}^k\|^2 \right).$$

- Update  $y_{ij}$  for each edge:

$$y_{ij}^{k+1} = \arg \min_{y_{ij}} \lambda_{ij}^T (x_i^{k+1} - y_{ij}) + \frac{\rho}{2} \|x_i^{k+1} - y_{ij}\|^2 + \mu_{ij}^T (y_{ij} - x_j^{k+1}) + \frac{\rho}{2} \|y_{ij} - x_j^{k+1}\|^2.$$

(b) **Dual Updates:**

- Update  $\lambda_{ij}$ :

$$\lambda_{ij}^{k+1} = \lambda_{ij}^k + \rho(x_i^{k+1} - y_{ij}^{k+1}).$$

- Update  $\mu_{ij}$ :

$$\mu_{ij}^{k+1} = \mu_{ij}^k + \rho(y_{ij}^{k+1} - x_j^{k+1}).$$

3. **Stopping Criterion:** Stop if constraints are satisfied within a tolerance.

## 10 Augmented Lagrangian Method (ALM) with Hard Thresholding

### 10.1 Problem Formulation

We aim to solve:

$$\min x^T A x - \mu^T x \tag{1}$$

subject to:

$$e^T x = 1, \quad x \geq 0, \quad \|x\|_0 \leq k. \tag{2}$$

### 10.2 Step 1: Reformulating the Problem for ALM

We introduce Lagrange multipliers and penalty terms to enforce constraints. Define:

- $c(x) = e^T x - 1$  (Equality constraint)
- $h(x) = -x$  (Non-negativity constraint)
- $\|x\|_0 \leq k$  (Cardinality constraint, handled separately)

The Augmented Lagrangian Function is:

$$\mathcal{L}_\rho(x, \lambda) = x^T A x - \mu^T x + \lambda(e^T x - 1) + \frac{\rho}{2}(e^T x - 1)^2. \tag{3}$$

The non-negativity constraint  $x \geq 0$  is imposed using projected updates, while the cardinality constraint  $\|x\|_0 \leq k$  is handled via hard thresholding.

### 10.3 Step 2: Iterative ALM Algorithm

#### 10.3.1 Initialization

Set  $x^0$ ,  $\lambda^0$ ,  $\rho > 0$ , and iteration counter  $k = 0$ .

#### 10.3.2 Step 1: Solve the Augmented Lagrangian Problem

At each iteration  $k$ , update  $x$ :

$$x^{k+1} = \arg \min_x \left( x^T A x - \mu^T x + \lambda^T (e^T x - 1) + \frac{\rho}{2} (e^T x - 1)^2 \right). \tag{4}$$

The gradient-based update is:

$$x^{k+1} \leftarrow x^k - \eta \nabla \mathcal{L}_\rho(x^k, \lambda^k), \tag{5}$$

where:

$$\nabla_x \mathcal{L}_\rho = 2Ax - \mu + \lambda e + \rho(e^T x - 1)e. \tag{6}$$

Apply projection onto the non-negative orthant:

$$x^{k+1} \leftarrow \max(0, x^{k+1}). \tag{7}$$



### 10.3.3 Step 2: Enforce Cardinality Constraint

Apply a hard thresholding operator:

$$x^{k+1} \leftarrow \mathcal{T}_k(x^{k+1}), \quad (8)$$

where  $\mathcal{T}_k(\cdot)$  retains the largest  $k$  elements and sets the rest to zero.

### 10.3.4 Step 3: Dual Update for Equality Constraint

Update Lagrange multiplier:

$$\lambda^{k+1} = \lambda^k + \rho(e^T x^{k+1} - 1). \quad (9)$$

### 10.3.5 Step 4: Increase Penalty Parameter

If needed, increase  $\rho$  to enforce constraint satisfaction:

$$\rho^{k+1} = \tau \rho^k, \quad \tau > 1. \quad (10)$$

### 10.3.6 Stopping Criteria

Stop when  $\|x^{k+1} - x^k\|$  and  $|e^T x - 1|$  are below a tolerance threshold.

## 10.4 Summary of ALM Algorithm with Hard Thresholding

- Initialize  $x, \lambda, \rho$ .
- Iterate until convergence:
  - Solve the quadratic problem for  $x$ .
  - Apply projection onto the non-negative orthant.
  - Apply hard thresholding to enforce sparsity.
  - Update Lagrange multiplier  $\lambda$ .
  - Adjust penalty parameter  $\rho$ .
- Return optimal  $x^*$ .

## 10.5 Why ALM Works Well Here

- Handles equality constraint explicitly via dual updates.
- Ensures non-negativity using projected gradient descent.
- Maintains sparsity via hard thresholding.

This hybrid ALM + hard thresholding approach efficiently solves the problem while enforcing all constraints.